

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-09

Sub-Millisecond High-Frequency Fraud Detection & AML Anomaly Engine

Domain: FinTech / Streaming Analytics / High-Performance Computing

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Deliver an ultra-low latency streaming transaction intelligence pipeline capable of scoring complex synthetic identity and layering fraud patterns within 15 milliseconds at 50,000 TPS.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] Sub-15ms P99 decisioning engine with sliding temporal feature stores.

[REQ-2] Dynamic graph link analysis identifying mule account rings and rapid money laundering chains.

[REQ-3] Adaptive behavioral biometric profiling (typing cadence, swipe dynamics, device fingerprints).

[REQ-4] High-throughput Apache Kafka / Redpanda event processing integration.

[REQ-5] Comprehensive regulatory AML SAR (Suspicious Activity Report) generation workflow.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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